

Nishad Singhi

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Education

Present Jul' 2024	TU Darmstadt PhD in Computer Science	Advisors: Dr. Anna Rohrbach, Dr. Marcus Rohrbach, Dr. Bernt Schiele
Mar' 2024 Nov' 2020	University of Tübingen MSc in Neural Information Processing	Advisors: Dr. Zeynep Akata, Dr. Wieland Brendel
Apr' 2020 Aug' 2016	Indian Institute of Technology (IIT) Delhi BTech in Electrical Engineering	Advisor: Dr. Sumeet Agarwal

Select Publications

* = Equal Contribution

- [P.1] **Think Twice, Act Once: Verifier-Guided ECoT for Reliable Embodied Agents**
N. Singhi, C. Bialas, M. Rohrbach, A. Rohrbach. *In Preparation.*
We improve OOD generalization in embodied agents by combining embodied chain-of-thought with verifier-guided selection.
- [C.1] **When To Solve, When To Verify: Compute-Optimal Problem Solving and Generative Verification for LLM Reasoning** COLM 2025
N. Singhi*, H. Bansal*, A. Hosseini*, A. Grover, K.W. Chang, M. Rohrbach, A. Rohrbach. [Paper] [Code]
We study how to best allocate test-time compute, finding that majority voting beats gen. verifier except at high inference costs.
- [J.1] **Centaur: a foundation model of human cognition** Nature [Preprint]
M. Binz, ..., N. Singhi, ..., E. Schulz.
We finetune an LLM on psychology data to obtain a foundation model of human cognition.
- [C.2] **Improving Intervention Efficacy via Concept Realignment in Concept Bottleneck Models** ECCV 2024
N. Singhi, K. Roth, J. Kim, Z. Akata. [Paper] [Talk] [Code]
We introduce a lightweight module that makes CBMs more efficient by minimizing the number of interventions required.
- [C.3] **CleanCLIP: Mitigating Data Poisoning Attacks in Multimodal Contrastive Learning** ICCV 2023 (Oral)
H. Bansal*, N. Singhi*, Y. Yang, F. Yin, A. Grover, K.W. Chang. [Paper] [Talk] [Code]
Best Paper Award at the Reliable and Trustworthy ML Workshop at ICLR 2023
Proposed a technique to remove backdoor attacks from CLIP by independently finetuning vision and language features.
- [C.4] **Toward a normative theory of (self-)management by goal-setting** CogSci 2023 (Oral)
N. Singhi, F. Mohnert, B. Prystawski, F. Lieder. [Paper] [Talk]
Diversity and Inclusion Award (10 recipients worldwide)
Built a tool to automatically decompose challenging problems into easier subgoals for humans, leading to 3x reduction in effort.
- [C.5] **Using Computational Models to Understand the Role and Nature of Valuation Bias in Gambles** CogSci 2023
N. Singhi, S. Agarwal, S. Mukherjee. [Paper]
Biases in valuing and attending to losses explain most individual differences in loss-averse decisions.

Select Experience

Present Jun 2025	DataComp Open-source Contributor / Advisor: Dr. Ludwig Schmidt Contributing to data curation and evaluation for software LLM agents.	Remote
Jul 2019 May 2019	University of California Los Angeles (UCLA) Summer Intern / Advisor: Dr. Tao Gao Developed a Deep-reinforcement learning pipeline to train multiple agents in a physical environment.	Los Angeles, CA

Honours and Awards

Prof. R. K. Mittal Award, 2017 Awarded to 2 freshmen (out of 850+) at IIT Delhi for academic performance.
IIT Delhi Merit Award, 2017 Conferred for being among the top 7% students of the batch at IIT Delhi.

Academic Service

Reviewer ICML 2025, CVPR 2025, ICCV CLVL Workshop 2025, TMLR